

## Communication Engineering

### Definitions :-

- (1) BANDWIDTH :- It is the range of frequencies over which information signal is transmitted.
- (2) Channel Bandwidth :- It is defined as the range of frequencies which a channel can pass through without any distortion.
- (3) Frequency spectrum :- It is the representation of a signal in freq. domain. It consists of amplitude and phase spectrum of signal.
- (4) Base Band signal :- It is defined as the <sup>message</sup> signal in the original form.  
Message signal may be in the form of text, sound, picture etc.  
Base band signal is also called as Modulating signal.

Modulation:- In modulation, some parameters of the carrier wave such as Amplitude, Frequency or phase is varied in accordance with the modulating signal, is called as Modulation.

Modulation is done at the transmitter side, and the resultant signal is called as Modulated Signal.

→ Modulated Signal is also called as Broad Band Signal.

# IEEE Frequency spectrum | EM spectrum and its

Applications :- (RADIO FREQUENCY SPECTRUM) :-

| <u>Frequency Band</u>                              | <u>Applications</u>                        |
|----------------------------------------------------|--------------------------------------------|
| 1. 30 Hz - 300 Hz<br>Extremely Low Freq. (ELF)     | In power Transmission                      |
| 2. 300 Hz - 3 KHz<br>Voice Frequencies (VF)        | Audio Applications                         |
| 3. 3 KHz - 30 KHz<br>Very Low Frequencies (VLF)    | Navy, Military Applications                |
| 4. 30 KHz - 300 KHz<br>Low Frequencies (LF)        | Aeronautical and Marine Navigation         |
| 5. 300 KHz - 3 MHz<br>Medium Frequencies (MF)      | AM Radio Broadcast                         |
| 6. 3 MHz - 30 MHz<br>High Frequencies (HF)         | Short Wave transmission                    |
| 7. 30 MHz - 300 MHz<br>Very High Frequencies (VHF) | TV Broadcasting                            |
| 8. 300 MHz - 3 GHz<br>Ultra High Frequencies (UHF) | Cellular phones,<br>Military Applications. |

9. 3 GHz - 30 GHz  
Super High Frequencies (SHF)

Radar and  
Satellite  
Communication

10. 30 GHz - 300 GHz  
Extreme High Frequencies (EHF)

Satellite and  
specialized  
Radars.

M. Jwp

Need for Modulation :- Modulation is Needed  
due to Following reasons :-

(1) To Reduce the height of Antenna :- For the  
transmission  
of radio signals, the Antenna height must be  
a multiple of  $\frac{\lambda}{4}$ .

→ Minimum Height of Antenna For  $F = 10 \text{ KHz}$  is -

$$\frac{\lambda}{4} = \frac{c}{f \times 4} = \frac{3 \times 10^8}{10 \times 10^3 \text{ Hz} \times 4} = 75 \text{ m}$$

→ Therefore we can see, As the frequency  
increases, the height of antenna decreases.

→ Minimum Height of Antenna For  $F = 1 \text{ MHz}$  -

$$\frac{\lambda}{4} = \frac{c}{f \times 4} = \frac{3 \times 10^8}{1 \times 10^6 \times 4} = 75 \text{ m}$$

② Modulation Avoids mixing of Signals :- IF baseband signals are transmitted by more than 1 transmitter then all the signals will get mixed up due to same bandwidth (0-20 kHz) and Receiver can't separate them from each other.

IF each baseband signal is used to modulate a different carrier, they won't get mixed up.

③ Increases the range of Communication :-

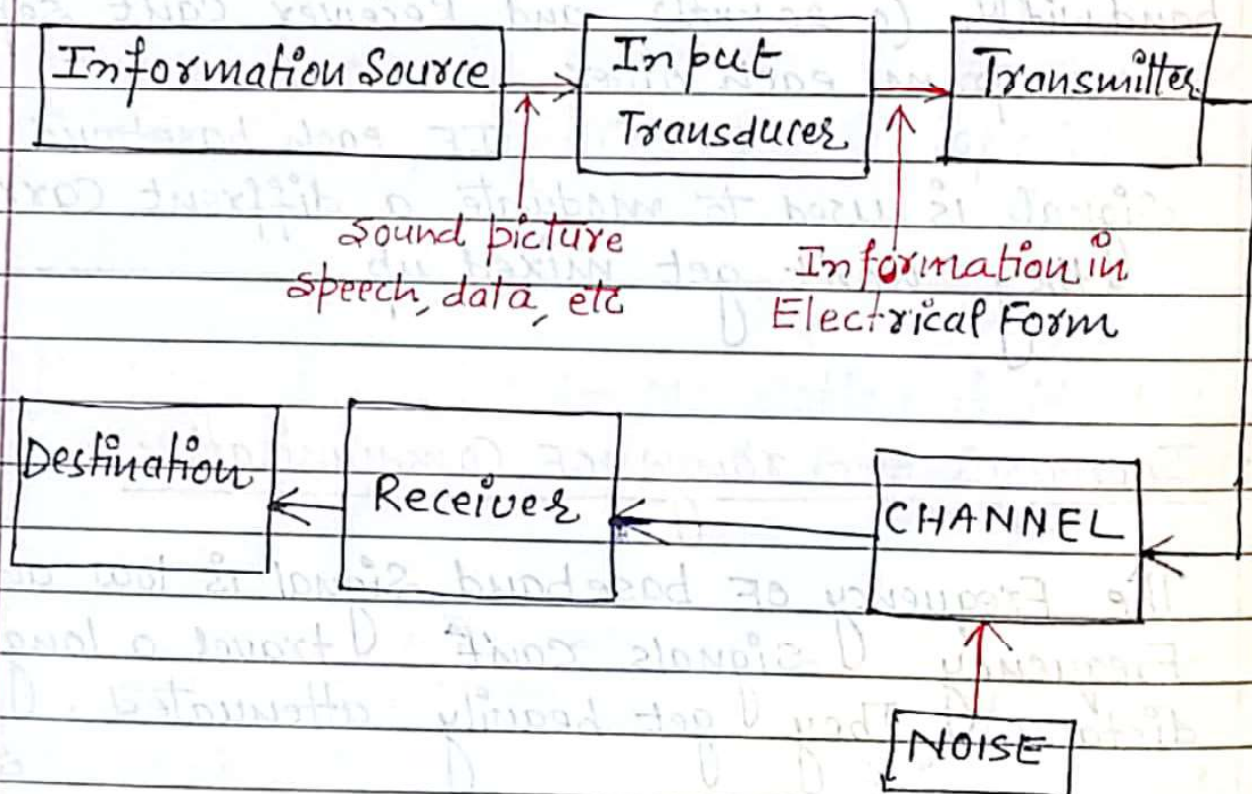
The Frequency of baseband signal is low and low frequency signals can't travel a long distance. They get heavily attenuated.

So by modulation we can increase the range of communication.

④ Makes Multiplexing possible :- Multiplexing is a process in which two or more signals can be transmitted over the same communication channel.

⑤ Improves Quality Reception :- With FM and digital communication techniques, the effect of noise is reduced to a great extent.

M. Imp

Block Diagram of Communication System :-

- ① Information Source :- Information source contains message. This message may be in the form of sound picture, speech, data, text etc. This information may be or may not be in electrical nature.

- ② Input Transducer :- Transducer is a device which transforms one form of energy into another form. The input

transducer is used to Convert the information or message, which is in non-electrical form, into a time varying electrical form.

- ③ Transmitter :- The transmitter processes the incoming information or message signal so as to make it suitable for transmission. Modulation is done at the transmitter.

- ④ CHANNEL :- channel is a medium through which the message travels from the transmitter to the receiver. In other words, the function of channel is to provide a physical connection between the transmitter and the receiver.

- ⑤ Noise :- Noise may interfere with signal at any point in the communication system. IF noise is present in the channel or at the input of receiver, it harms a lot.

- ⑥ Receiver :- The function of receiver is to receive the signal (modulated) and to extract the message, which is in electrical form, from the carrier signal that means De-Modulation is done at the receiver.

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⑥ Destination :- This is the Final stage which is used to Convert an electrical message signal ~~its~~ into its original Form.

For this Inverse transducer is used.

# Cellular Networks

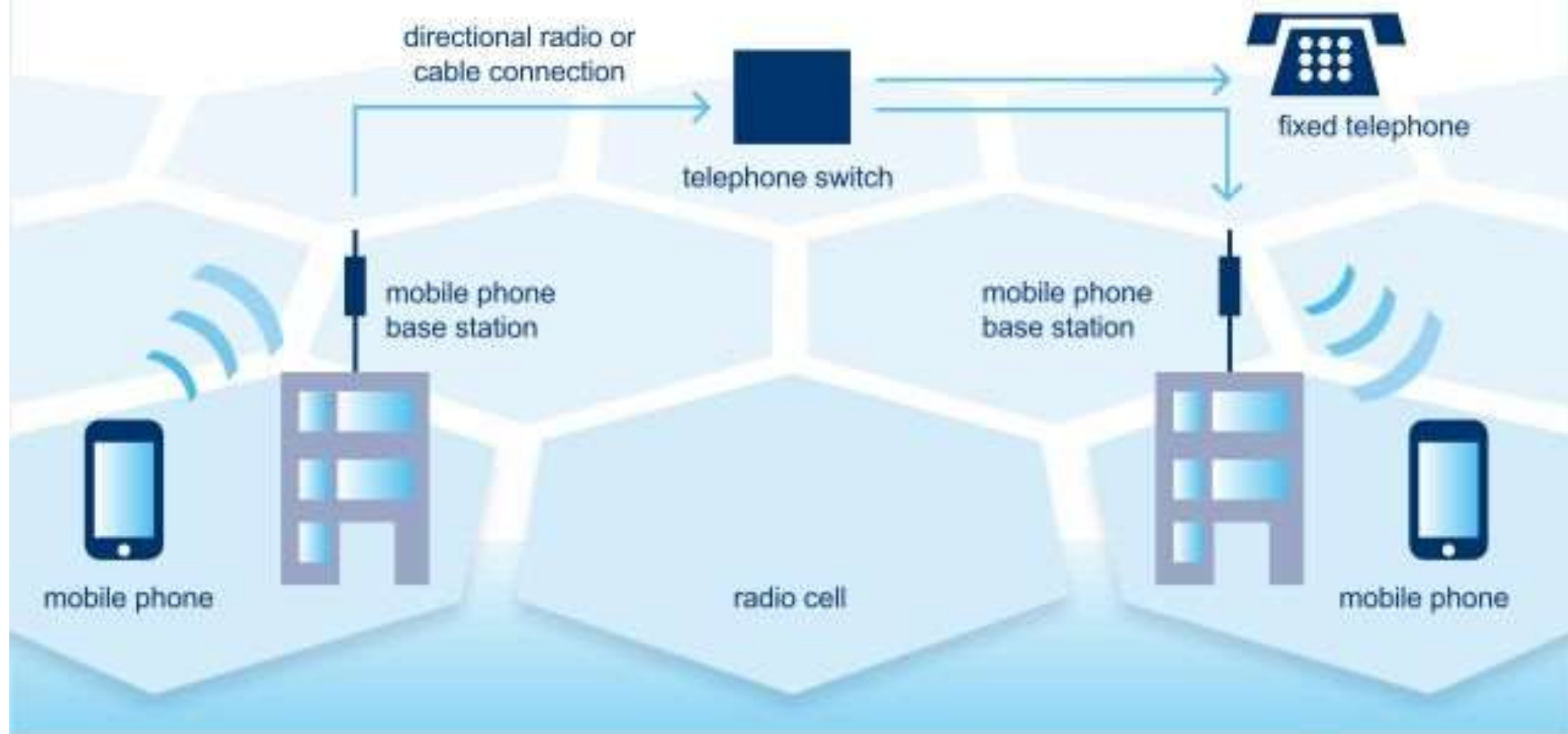
# Outline

- An overview
- Different generations in cellular communication systems
- Introduction to 5G and 6G

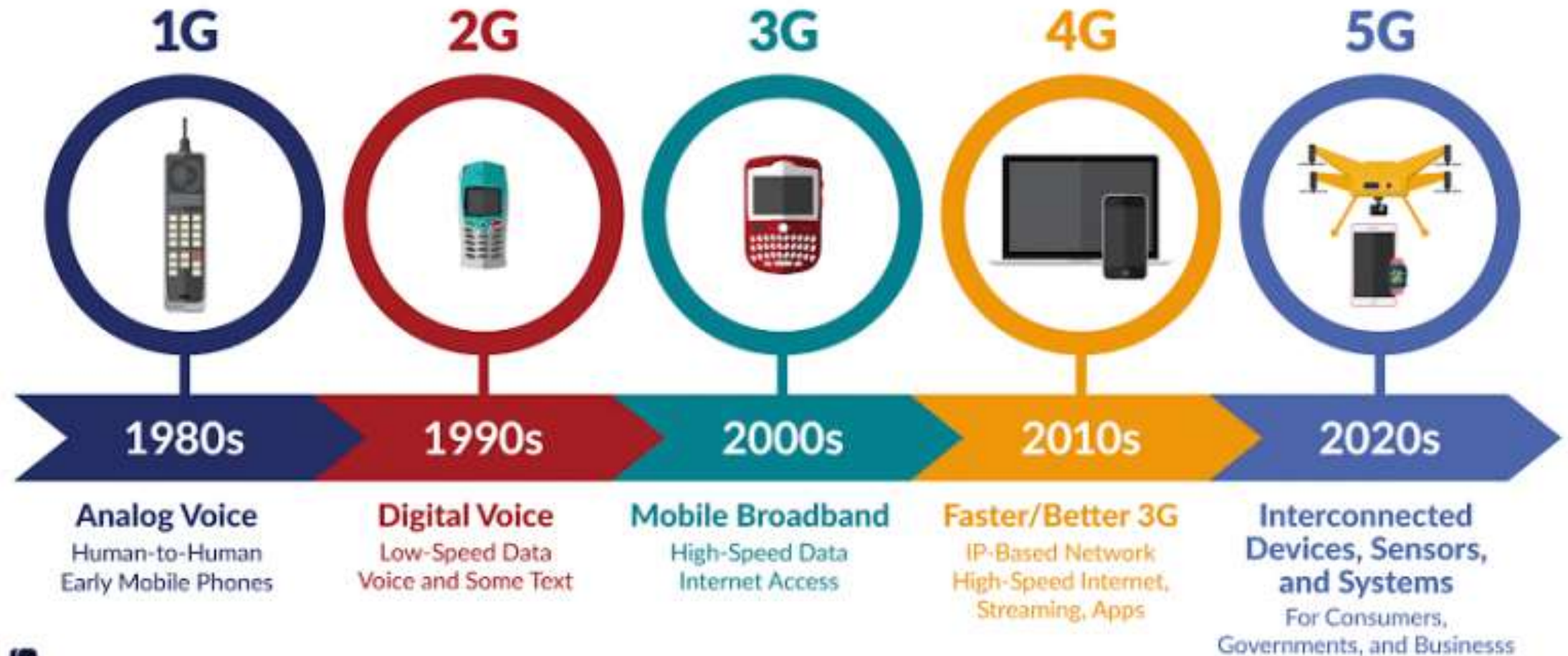
# An Overview

- A cellular network (or mobile network as they are also known) is a wireless communication system that divides a geographic area into sections called cells.
- Each cell is served by a base station and various other components working together enabling wireless communication to mobile and IoT (Internet of Things) devices.
- Cell networks allow mobile devices to connect to the internet, make phone calls and send / receive texts or cellular data.
- They do this by switching frequencies seamlessly as users move across different cells.

## How does a mobile phone network work?



# Different Generations in Cellular Networks



# 1G Networks (1<sup>st</sup> Generation)

- Mobile telecommunications which was first introduced in 1980s and completed in early 1990s.
- Speed was up to 2.4kbps.
- The Operating frequency of 1G is 800 MHz and 900 MHz.
- First-time calling was introduced in mobile systems.
- It used analog signals.
- The coverage area was small.
- No roaming support between various operators.
- Only the voice can be transmitted in 1G and its quality is very low.
- Mobile Phones could not have a long life in battery usage and were very big which was very difficult to carry.

# 2G Networks (2<sup>nd</sup> Generation)

- Shifted from analog to digital.
- To have advancement over the 1G, the GSM (Global System for Mobile Communication) was introduced.
- In the current technology, every system has the basis of GSM. This generation is responsible for the upgradation from voice calls to MMS (Multimedia Messaging System) and SMS (Short Message Systems).
- The operating frequency of 2G is 1.8 GHz.
- The data speed of 2G is nearly up to 64 Kbps.
- The Voice quality also improved from 1G, it supports both voice and data.

# 2G Networks (2<sup>nd</sup> Generation)

## (contd..)

- It needs strong network coverage for data transmission.
- Even though it helps in sending messages, it cannot handle videos for transmission.

# 2.5G & 2.75G Networks

- **2.5G** came after **2G** which used the concept of GPRS.
- (**General Packet Radio Service**, is a mobile data standard that provides internet access for 2G and 3G cellular networks). High data rate of 171kbps. Using the packet switching technique, the data is transferred, by dividing the data into small packets for an easy mode of transmission.
- Then came **2.75G** or **EDGE (Enhanced Data Rates for GSM)** with an increased data rate of up to 473.6 Kbps. Technology used is CDMA with a data rate of 384 Kbps. which was faster in providing services than 2.5G.
- It supports Email transmission and Internet browsing and also features cameras in cellular phones.

# 3G Networks (3<sup>rd</sup> Generation)

- It comes with data transfer technology such as UMTS (Universal Mobile Terrestrials System) for use in smartphones.
- With this feature, we can have the video calling option and play games online on cellular phones.
- Better system and capacity.
- Offers high-speed wireless internet.
- It supports both voice and data with a data rate of 3.1 Mbps.

# 3G Networks (3<sup>rd</sup> Generation)

## (contd..)

- It has high security and international roaming.
- The power consumption of using 3G technology is high.
- The time taken to download via 3G is faster and more expensive to buy.

# 4G Networks (4<sup>th</sup> Generation)

- LTE (Long term evaluation) & Wireless Fidelity (Wi-Fi) data technology are used.
- It provides faster data rate to upload and download data.
- Vo-LTE (Voice over LTE) is for both voice and the internet.
- Freedom and flexibility to select any desired service with reasonable Quality of Service.
- High usability.
- Supports multimedia service at a low transmission cost.
- HD Quality Streaming.
- Speed:-100mbps.

# 4G Networks (4<sup>th</sup> Generation)

## (contd..)

- The Data rate provides a capacity of 100Mbps to 1Gbps and is highly securable by using the Encryption Technique.
- The Internet Service provided by 4G is Ultra-Broadband.
- It provides high speed and global mobility using the internet and International calls.
- It provides costlier hardware and infrastructure setup.
- Due to the high data rate speed, the battery usage is less.

# 5G Networks (5<sup>th</sup> Generation)

- Higher data rates.
- Connectivity is more fast and more secure,
- .Massive network capacity.
- It is 30 times faster than 4G.
- Faster loading speeds of real time applications such as maps & weather
- The ability to have multiple people in video conferencing calls
- Ability to stream high-definition (HD) .

# Beyond 5G Networks

- Beyond 5G" (B5G) primarily refers to the research and early development stages of **6G technology**, which aims to expand on 5G's capabilities to enable a hyper-connected society where physical and digital worlds will merge.
- Ultra-high data rates, integration with artificial intelligence (AI),
- Seamless connectivity across all locations, including remote areas, oceans, and air.
- Enables advanced Extended Reality (XR) applications, including holographic communication.
- Focus on reducing power consumption in both devices and network infrastructure.

# Sensors & Transducers

# Sensors

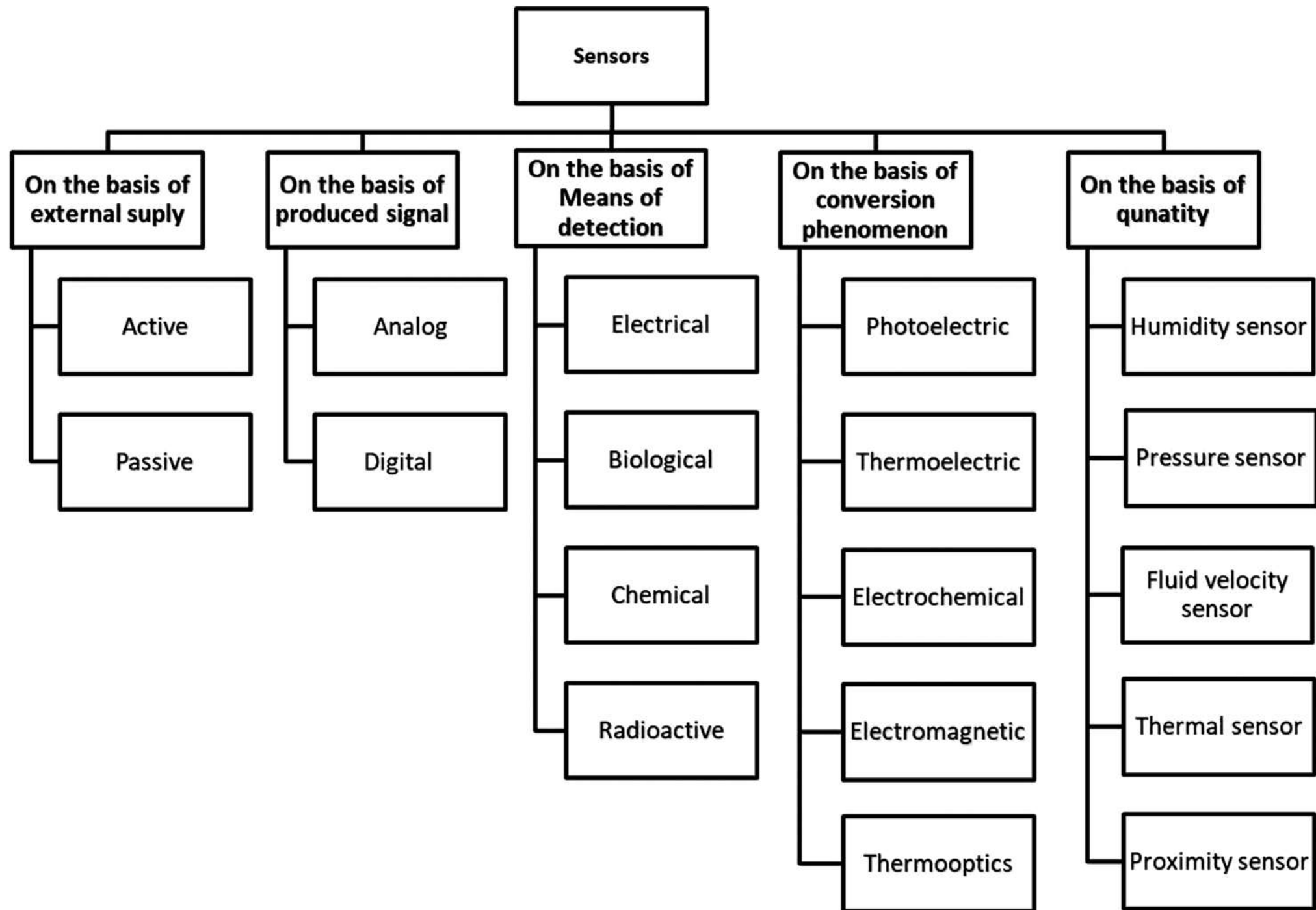
- Devices that detect and measure physical properties or inputs from the physical environment.
- Sensors can detect various parameters such as temperature, pressure, light, humidity, motion, and many more.

Example: **Thermistors**: They measure temperature by measuring the change in resistance of the electric current, **Photosensors**: detect the presence of visible light, infrared transmission, or ultraviolet energy

# Difference between Transducer and Sensor

The differences between transducer and sensor have been discussed in this table:

| Transducers                                                               | Sensors                                                                                                                                                     |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Transducers are devices that convert the energy of one form into another. | Sensors are devices that convert physical quantities into either a readable form or into electrical signals which are then used with transducers in tandem. |
| Examples are microphones, LVDT, LED, loudspeakers, etc.                   | Examples are pressure sensors, motion sensors, etc.                                                                                                         |



# Few Real-life Applications of Sensors

- Medical devices like digital thermometers and MRI machines that monitor patient temperatures.
- Aerospace applications: They also measure cabin pressure, altitude, and hydraulic system performance.
- Smart devices: Sensors also enable touchless gestures in smartphones and voice assistants.
- Cameras and imaging systems: Enable autofocus, HDR, and ambient light adjustments in smartphones and digital cameras.
- Environmental monitoring: Measure air pollution and gas levels in smart city applications.
- Proximity sensors automatically turn off smartphone screens when placed near the ear.

A modern smartphone typically has **10 to 15 different types of sensors**, including core ones like the accelerometer, gyroscope, magnetometer (compass), proximity, and ambient light sensors, plus advanced ones like fingerprint scanners, Time-of-Flight (ToF) sensors, SpO2 sensors, and barometers for altitude, enabling features from screen rotation to advanced health tracking.

### **Common Sensors & Their Functions:**

- Proximity Sensor: Turns off the screen when you hold the phone to your ear during calls.
- Accelerometer: Detects motion, acceleration, and device orientation (portrait/landscape).
- Gyroscope: Measures rotational movement (pitch, roll, yaw) for better orientation and gaming.
- Magnetometer (Digital Compass): Senses magnetic fields to help with navigation and compass apps.
- Ambient Light Sensor: Adjusts screen brightness automatically based on surroundings.
- Fingerprint Sensor: For secure biometric authentication.

- Camera/Image Sensor**: Captures photos and videos, with some adding depth sensing (ToF).
- Barometer**: Measures air pressure to determine altitude (e.g., for fitness tracking).
- Microphone**: Detects sound for calls, voice commands, and noise cancellation.
- Pedometer**: Counts steps and tracks activity using motion data

#### **Premium & Advanced Sensors :**

**Time-of-Flight (ToF) Sensor** : Measures distance for enhanced portrait mode (bokeh).

**Heart Rate & SpO2 Sensors** : Found in some health-focused phones for vital signs.

**Thermometer** : Monitors device or environmental temperature.

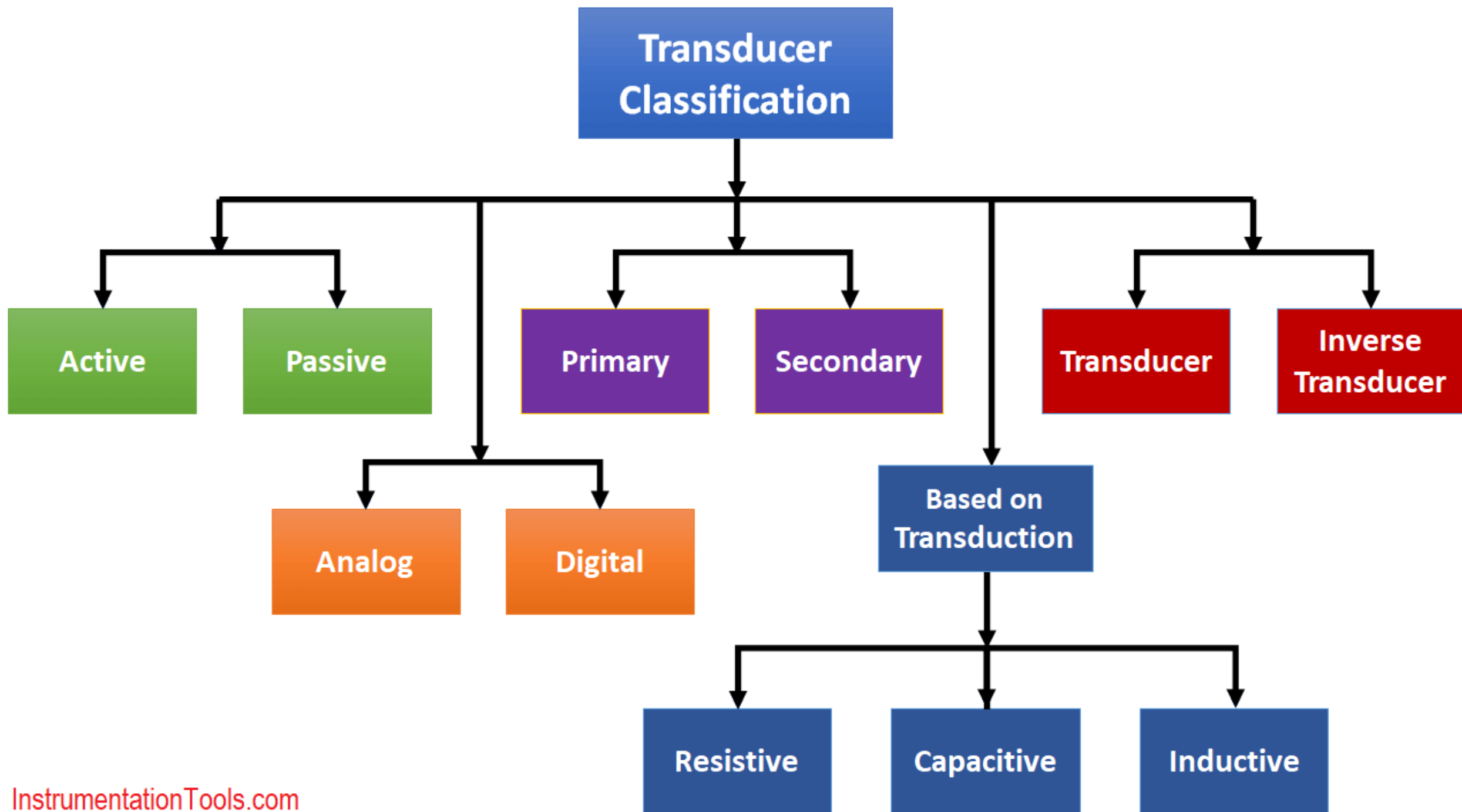
**Humidity Sensor** : Measures moisture in the air .

**IR Blaster** : Allows the phone to control other devices like TVs .

# Transducers

- Transducers are a **type of sensor** that not only detects physical quantities but also converts one form of energy into another.
- They act as both sensors and actuators, allowing **the measurement of a physical parameter and the conversion of that measurement into a usable form (most common form is electrical energy).**

For example, a **microphone** is a transducer that converts sound waves into electrical signals



# Few Real-life Applications of Transducers

- **Microphones:** Microphones convert sound waves into electrical signals. They are used in a wide variety of applications, such as recording studios, public address systems, and hearing aids.
- **Speakers:** Speakers convert electrical signals into sound waves. They are used in a wide variety of applications, such as home stereos, car stereos, and public address systems.
- **Hydrophones:** Hydrophones convert sound waves in water into electrical signals. They are used in a variety of applications, such as underwater communication, sonar, and oceanography.
- **Geophones:** Geophones convert ground vibrations into electrical signals. They are used in a variety of applications, such as seismology, oil exploration, and mining.
- **Accelerometers:** Accelerometers convert acceleration into electrical signals. They are used in a variety of applications, such as navigation, robotics, and medical devices.

# Actuators

- Actuators are devices that convert electrical signals into physical action or movement.
- Unlike sensors that measure physical properties, **actuators generate a physical response based on the signals they receive.**
- Examples of actuators include **motors, solenoids, valves, and relays.**

# Few Real-life Applications of Actuators

1. In factories , actuators run machines, move conveyor belts, and control robot arms. They help make products faster, more accurately, and with less human error.
2. In aeroplanes, actuators control things like landing gear, flaps, and engines. They must work perfectly to keep flights safe and smooth.
3. For robots, actuators let robots move and do tasks like picking things up, welding, or painting. The type of actuator used affects how well the robot works.
4. In vehicles, actuators help with steering, braking, and engine control. They make driving safer, easier as well as more fuel-efficient.
5. In heating and cooling systems, actuators control airflow, fans, and temperature settings to keep rooms comfortable.
6. In Smart Homes, actuators are used in smart homes to open blinds, lock doors, and turn on lights automatically. This generally makes life easier and saves energy.
7. In healthcare, actuators are found in things like artificial limbs, surgical robots, and testing machines. They help doctors as well as improve patient care.

# Introduction to IoT Architecture

# What is IoT?

- The Internet of Things (IoT) is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction.
- The Internet of Things is the concept of connecting any device (so long as it has an on/off switch) to the Internet and to other connected devices.

# Applications of IoT

- Smart home ☐
- Smart Wearables
- Connected Cars
- Smart cities
- IoT in agriculture
- Smart retail
- IoT in healthcare
- IoT in Poultry and Farming

## 4 Stage IoT Architecture

Data Flow



### Application Layer

Smart Applications and Management

Smart  
Application

### Data Processing Layer

Processing Unit – Data Analytics/ Decision Unit

Process  
Information

### Network Layer

Internet Gateway/ Network Gateway – Network Technology

Data  
Transmission

### Sensing Layer

Physical Object – Sensor and Actuator

Data  
Gathering

# 1. Perception / Sensing Layer

- The first layer of any IoT system involves “things” or endpoint devices that serve as a connection between the physical and the digital worlds.
- Perception refers to the **physical layer**, which includes sensors and actuators that are capable of collecting, accepting, and processing data over the network.
- Sensors and actuators can be connected either wirelessly or via wired connections.

## 2. Network Layer

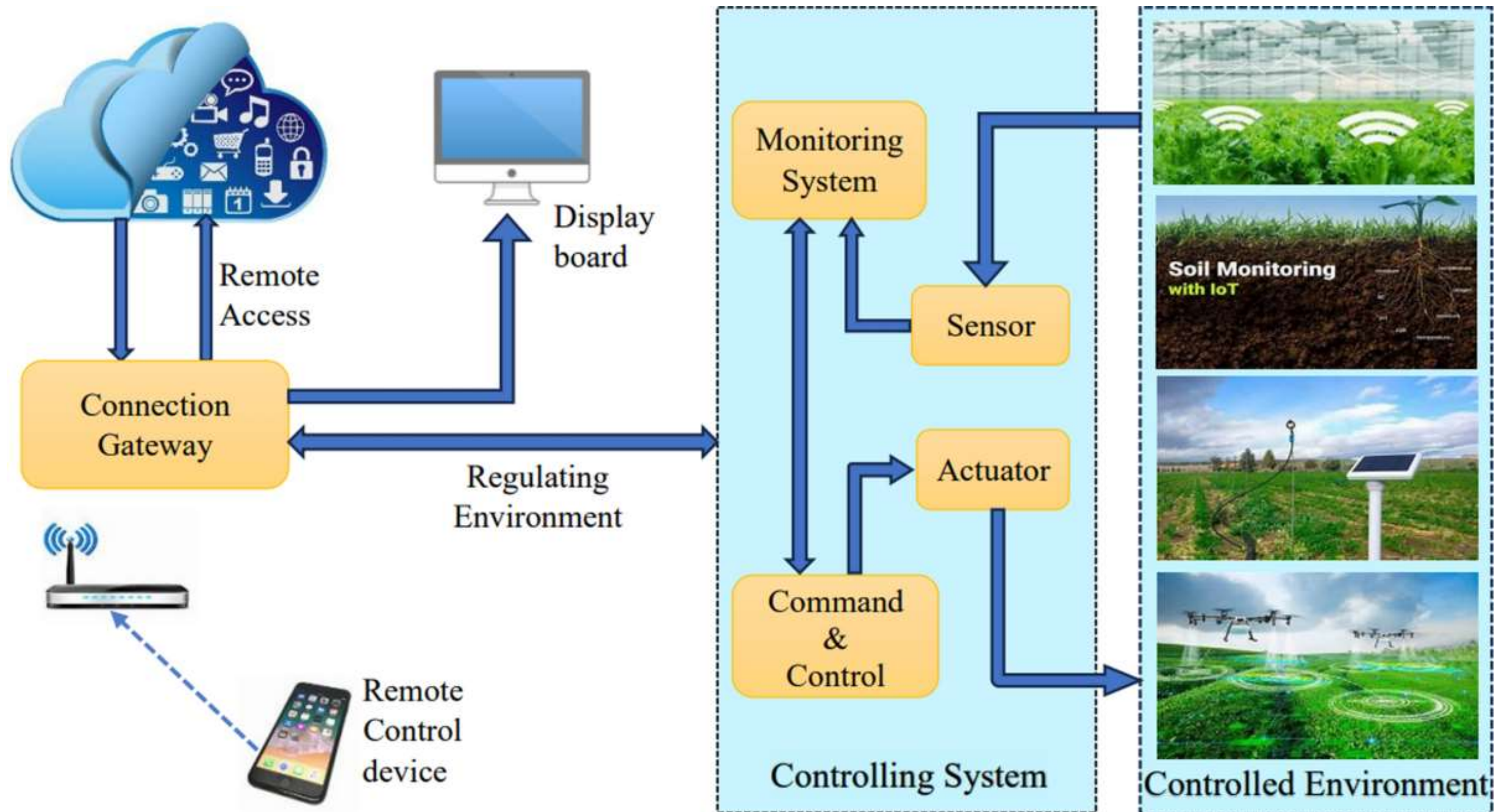
- Network layers provide an overview of how data is moved throughout the application.
- Transmits collected sensor data to processing platforms securely using Data Acquiring Systems (DAS) & Internet/Network gateways.
- Supports device-to-device and device-to-server communication.
- Handles addressing, routing and data forwarding.
- Technologies - Wi-Fi, Bluetooth, Zigbee, LoRaWAN (Long Range Wide Area Network - a low-power wireless protocol for Internet of Things (IoT) devices )
- Uses Ethernet and satellite networks, Supporting Devices, Routers and switches

# 3. Data Processing Layer

- The processing layer is the brain of the IoT ecosystem.
- Data is analyzed, pre-processed, and stored here before being sent to the data center, where it is accessed by software applications that monitor and manage the data as well as prepare further actions.
- Uses IoT cloud platforms, Data lakes and warehouses
- Utilizes Stream processing and machine learning engines
- Generates predictive alerts and reports; Anomaly detection signals; Decision rules for automation

## 4. Application Layer

- User interaction takes place at the application layer, which delivers application-specific services to the user.
- Provides interfaces to monitor and control IoT devices remotely.
- Leads to Remote device management, Real-time condition monitoring, Integration with enterprise applications
- Displays visual analytics through dashboards and charts.
- Triggers automated actions based on processed data

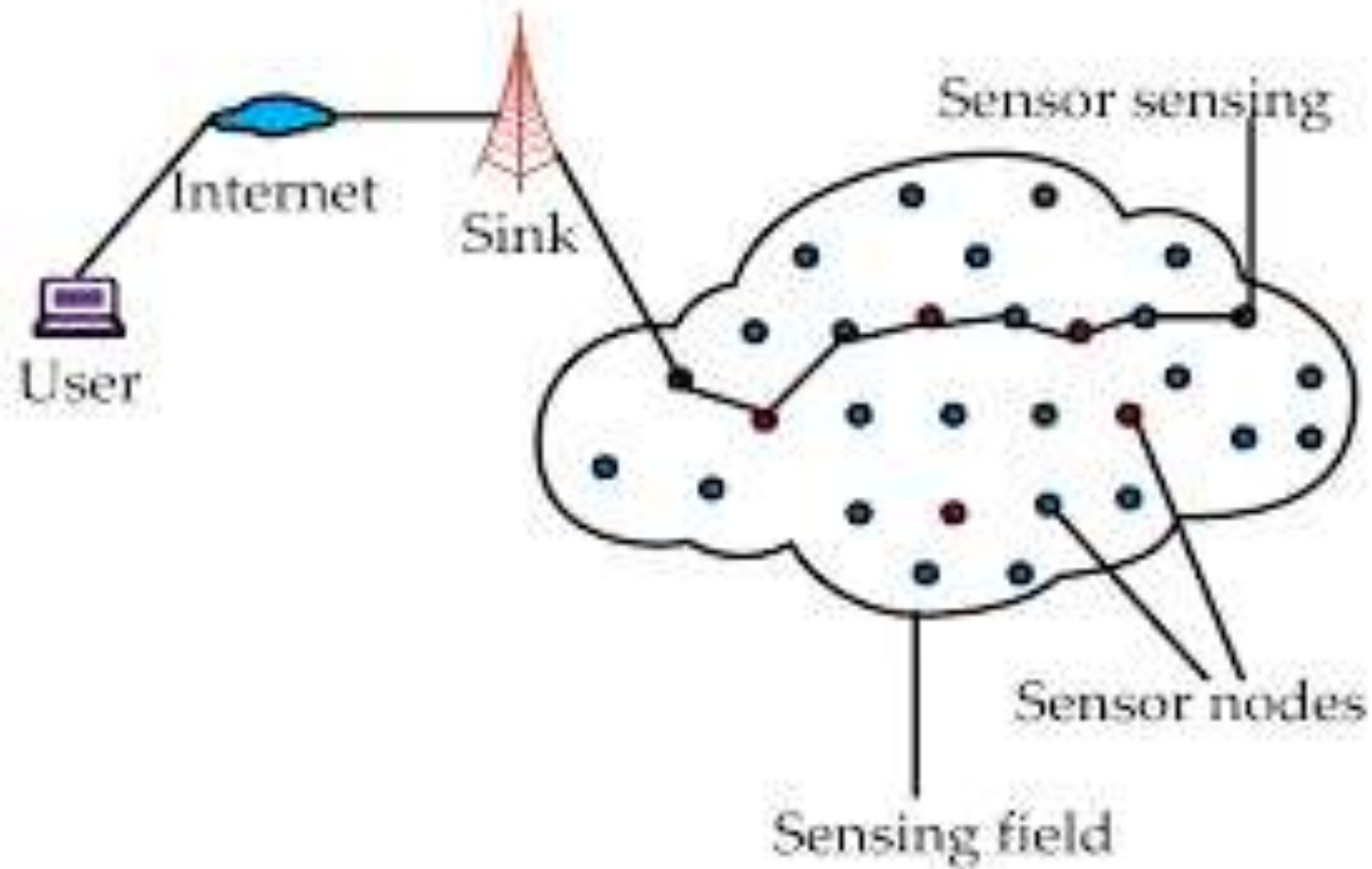


# Wireless Sensor Networks (WSN)

# An Introduction

- Wireless Sensor Networks (WSNs) is wireless communication, allowing sensors to transmit and receive information without the limitations of wired connections.
- Wireless sensor networks (WSNs) systems are made up of small, wireless sensors distributed across an area to collect and share data about the environment.
- These networks play a vital role in gathering real-time information and are commonly used in smart cities, industrial automation, environmental monitoring, and healthcare.
- As technology advances, WSNs have become a key component of the Internet of Things (IoT), connecting devices and streamlining operations.

# Architecture of WSN



# Key Components of WSN

- **Sensor Nodes (Motes):** Tiny devices with sensors (transducers), a processor, memory, transceiver (radio), and power unit (battery/energy harvester).
- **Sink Node/Gateway:** Collects data from sensor nodes and forwards it to the internet or a management system.
- **Application Platform:** Where data is analyzed, visualized, and acted

# How WSN works?

- 1. Sensing:** A sensor node's transducer detects a physical parameter (e.g., heat, light, motion) and converts it into an electrical signal.
- 2.Signal Processing:** An Analog-to-Digital Converter (ADC) digitizes the signal, and a microcontroller processes it, often putting the node to sleep to save power.
- 3.Communication (Multi-Hop):** Nodes send data wirelessly (using protocols like Zigbee, LoRa). If a node can't reach the sink directly, intermediate nodes act as relays, passing data across the network (mesh/multi-hop).
- 4.Data Aggregation:** The sink node receives data from many nodes, possibly performing initial aggregation.
- 5.Data Forwarding:** The gateway sends aggregated data to the internet or a backend system for users to access

# Applications of WSN

## **Military applications**

- Battlefield surveillance and monitoring
- Intelligence, facility control, and communication systems
- Detection of chemical or biological attacks

## **Environmental and area monitoring**

- Monitoring air pollution, temperature, humidity, and noise levels
- Forest fire and landslide detection
- Water quality monitoring
- Habitat and wildlife monitoring
- Radiation monitoring, Disaster prevention

# Applications of WSN (contd...)

## Healthcare applications

- Remote and continuous patient monitoring and diagnostics
- Monitoring of physiological information
- Drug administration and management
- Tracking patients or staff within a facility

## Smart environments and infrastructure

- **Smart homes:** Automation and remote monitoring
- **Smart cities:** Urban surveillance, traffic control, and smart grid deployment
- **Industrial automation:** Data collection, process monitoring, and control

# Applications of WSN (contd..)

## Agriculture

- Precision agriculture for monitoring soil conditions, temperature, and sunlight
- Automated irrigation systems based on real-time data
- Tracking crops and livestock